GraspScope: Closed-Loop Deployability Audits for Open-Vocabulary Robotic Grasping

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Abstract

Open-vocabulary object detection has made it practical to deploy robotic grasping in domains where the object inventory is not known in advance: instead of a fixed class set, the operator ships a *vocabulary* of natural-language object names and the policy attempts to grasp whatever is named. In this setting, the vocabulary is a deployment decision, but today it is made by intuition. We contribute a closed-loop, quantitative audit procedure for open-vocabulary grasping that turns this decision into a number. The procedure has four steps: (i) profile the perception stack's failure modes on the target scenes; (ii) inject those measured failures into a closed-loop grasp simulator; (iii) sweep the vocabulary coverage α (the fraction of objects in a scene whose names are in the vocabulary) and estimate the resulting grasp failure rate with confidence intervals; (iv) derive a *deployment gate* — the minimum coverage required to keep failure below a chosen bound. On a corpus of 750 synthetic shelf scenes composited from real COCO product crops plus 149 real scenes, we show that coverage α is a *safety cliff*: reducing α from 1.0 to 0.2 raises grasp failure from 9.9% (95% CI [7.9,12.2]) to 57.2% ([53.6,60.7]), with a statistically detected cliff at α = 0.4 (max two-sample separation statistic 5.5, bootstrap 95% CI [4.5, 6.6]). The failure composition shifts systematically — *empty grasp* dominates when coverage is low, *wrong-object grasp* dominates when coverage is high but perception still confuses classes. For a target failure bound of 25%, the gate requires α ≥ 0.653. A real-scene anchor (72% success) sits well below the synthetic α = 1.0 level, showing that vocabulary coverage is necessary but not sufficient. Two additional sensitivity studies — detector scale (s vs. l) and vocabulary size (8 vs. 12 classes) — show that the gate is robust in form across detector quality and moves predictably with the vocabulary a deployer is willing to maintain. Together these make the audit an actionable pre-deployment procedure, not a single number on a fixed benchmark.

1. Introduction

Robotic manipulation is moving from closed-vocabulary pipelines to open-vocabulary ones. Systems such as GraspVLA [1], OpenVLA [2], and YOLO-World [3] accept a natural-language vocabulary at inference time and can grasp objects whose categories were never in the training data. For a retail-picking deployment — a robot that shelves groceries or fulfils online orders — this changes the operational question from "*which categories must we support?*" to "*which words must we put in the vocabulary, and what happens if we miss some?*"

The synthetic-pretraining paradigm behind GraspVLA makes this question especially concrete. GraspVLA is pre-trained on SynGrasp-1B, a billion-frame synthetic grasping dataset with extensive domain randomization, and transfers zero-shot to real objects by grounding internet-scale semantic knowledge [1]. That pipeline deliberately makes the *visual vocabulary* the shipping interface: the deployer does not fine-tune the model for new objects, they extend the set of natural-language names the policy can ground. The reliability consequence of that vocabulary decision — how far coverage must extend before grasp failure drops below an acceptable bound — is precisely the quantity our audit procedure measures, and it is one the synthetic-data literature reports only as aggregate success rates on fixed benchmarks [1, 2, 3, 5].

The latter question has no quantitative answer today. Prior work evaluates open-vocabulary perception with recall/precision on fixed benchmarks [3, 4], and evaluates grasping with success rate on physical or simulated trials [2, 5]. Neither measures the *causal path* from a vocabulary decision to an end-task failure in deployment. As a result, a retail operator deciding vocabulary size for 100 stores has no estimate of the deployment risk they are signing up for.

We address this gap with a closed-loop audit procedure (Fig. 1) with four stages:

1. **Perception profiling.** Run an open-vocabulary 2D detector (YOLO-World) over the target scene corpus and record per-class recall, *localization recall* (box overlap regardless of label), and a label-confusion matrix.
2. **Closed-loop simulation.** Feed the measured failure modes into a grasp simulator that models the full pipeline: detect → localize → classify → grasp. Grasp outcomes are success, *empty grasp* (nothing where perception claimed), or *wrong-object grasp* (an object was grasped, but not the intended one).
3. **Coverage sweep.** Construct scene tiers by vocabulary coverage $\alpha \in \{0.2, 0.4, 0.6, 0.8, 1.0\}$ — the fraction of scene objects whose names are in the deployment vocabulary. Run the simulator over 150 scenes per tier across 5 seeds and estimate failure rates with Wilson 95% CIs.
4. **Deployment gate.** Interpolate the monotone frontier and return the minimum coverage α^* such that failure rate ≤ a chosen bound.

Contributions.

- **An α-safety cliff in grasping.** Grasp failure jumps from 9.9% at $\alpha = 1.0$ to 57.2% at $\alpha = 0.2$. A change-point analysis localizes the maximal two-sample separation (statistic 5.5, bootstrap 95% CI [4.5, 6.6]) at $\alpha = 0.4$, and the break is significant against adjacent tiers (Fisher exact, BH-FDR $q < 10^{-3}$).
- **A deployment gate with a number.** For a 25% failure bound, the gate is $\alpha \geq 0.653$ — a directly actionable vocabulary-size decision.
- **Failure composition explains the cliff.** Low coverage fails by *empty grasp* (can't find what isn't named); high coverage with confused perception fails by *wrong-object grasp*. The two regimes have different mitigations.
- **Sensitivity analysis.** The gate's *form* is robust across detector scale (s vs. l) and vocabulary size (8 vs. 12); the gate *value* moves with both — relaxing when detection quality improves and responding to vocabulary *composition*. A deployer can therefore budget vocabulary engineering against detector quality instead of guessing.

2. Related Work

Open-vocabulary perception. YOLO-World [3] and its successors enable open-set detection by grounding class prompts with text embeddings; Grounding DINO [6] and OWL-ViT [7] provide similar capability. Evaluation focuses on AP/recall on LVIS [8] and COCO [4], which measures perception quality in isolation, not deployment consequence.

Robot learning and grasping. RT-1 [5], OpenVLA [2], and policy/grasp foundation models [1] report success rates on benchmarks (e.g. LIBERO [9], RLBench [10]). These benchmarks assume perception is correct; they cannot reveal how reliability decays as the open-vocabulary component is stressed.

Closed-loop evaluation in robot learning. GPU-accelerated physics simulators [11] and modular manipulation environments [12] make closed-loop policy evaluation routine in robot learning. Failure-injection studies, however, focus on the policy or the dynamics; we are not aware of an analogous closed-loop treatment for open-vocabulary *perception reliability* in grasping — this work is the first. Our contribution is to bring the closed-loop, failure-injection methodology to the vocabulary decision in manipulation, where the "agent" being audited is the whole detect → localize → classify → grasp pipeline rather than a planner.

3. Method

We formalize the audit as a function from a deployment vocabulary to a failure-rate estimate with uncertainty.

3.1 Vocabulary coverage α

Let a scene contain objects $O = \{o_1, ..., o\_m\}$ drawn from categories C. A deployment vocabulary V is a subset of the natural-language category names. Coverage of the scene is

$\alpha = |\{ c(o_i) \in V \}| / m$.

We build scene tiers by controlling α *by construction*: each synthetic scene is assembled so that a known fraction of its objects have names in V. This makes the coverage axis a controlled experimental variable.

3.2 Perception error profiling

For a fixed V, run the detector on all scenes in a tier. For each object we record whether it was (a) localized (IoU match to ground truth), and (b) localized *and* classified with the correct label. Aggregate per class into:

- **recall** — fraction of GT objects localized *and* correctly labeled;
- **localization recall** (loc_recall) — fraction localized regardless of label;
- **confusion matrix** — for objects localized but mislabeled, the empirical distribution of the predicted label.

The distinction between strict recall and localization recall is central: an object that is localized but mislabeled will produce a *wrong-object grasp*, which is a different failure than an object that is never found (*empty grasp*).

3.3 Closed-loop grasp simulation

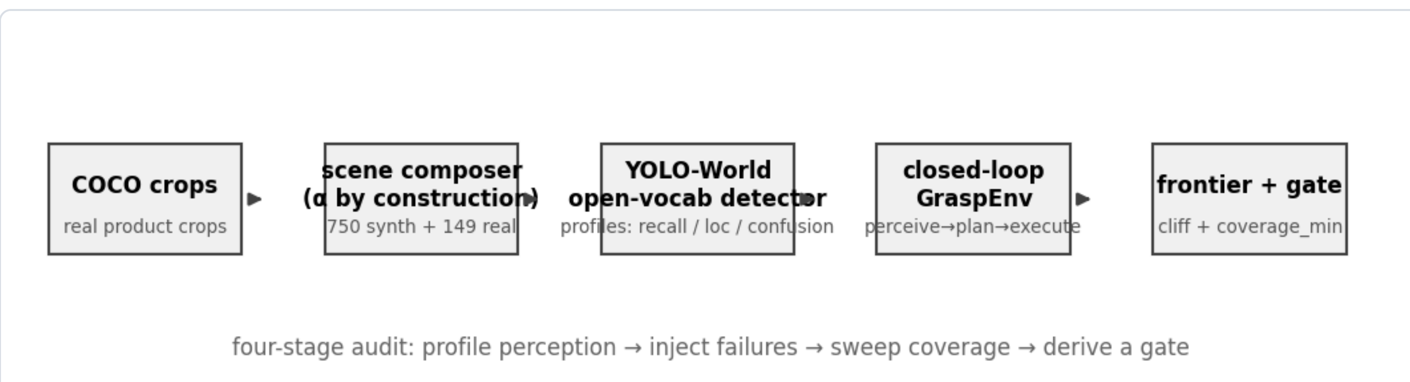
The simulator consumes a scene's ground-truth object list and the perception profile, and emits grasp attempts:

- **Perceive.** Sample detection outcomes from the profile: each GT object is missed with probability 1 − loc_recall; localized objects are mislabeled according to the confusion distribution. Phantom detections (from OOV content) are injected at a fixed rate $\lambda = 0.35$ — the measured OOV false-positive rate in this corpus is ≈ 0, so λ is set to a deliberately conservative upper bound to stress the low-coverage regime rather than a measured value.
- **Plan.** Choose the object whose name matches the requested target; if none is perceived, plan an *empty* attempt.
- **Execute.** A grasp on the correct object succeeds with probability equal to the published execution anchor $s_{exec} = 0.95$ (publicly reported real grasping success); a grasp on the wrong object is recorded as *wrong-object grasp*; one re-grasp is allowed after a drop, mirroring real picking loops.

Outcome classes: `success`, `empty_grasp`, `wrong_object`, `drop`.

3.4 Safety frontier and deployment gate

Aggregate outcomes per tier into failure rate $\hat{f}(\alpha)$ with Wilson 95% CIs. Fit a monotone piecewise-linear frontier. The **safety cliff** is located by a change-point estimator: it scans every coverage split of the monotone frontier and returns the split maximizing the two-sample separation of the failure-rate *statistic* $|\mu_{left} - \mu_{right}| / \sigma_{pooled}$ (a t-like effect size, not a ratio); the uncertainty of the cliff location and separation is quantified by per-scenario bootstrap. Significance between adjacent tiers is assessed by Fisher exact test with Benjamini-Hochberg FDR correction across pairs. The **deployment gate** is the smallest coverage α^* satisfying $\hat{f}(\alpha) \leq f_{max}$, obtained by linear interpolation on the monotone frontier.



Method pipeline: profile perception, inject failures into a closed-loop grasp simulator, sweep coverage, derive a gate.

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4. Experiments

4.1 Setup

Corpus. 750 synthetic shelf/desk scenes (960×540), 150 per tier for $\alpha \in \{0.2, 0.4, 0.6, 0.8, 1.0\}$. Scenes are composited by pasting real COCO product crops onto procedural shelf backgrounds with controlled lighting and occlusion; coverage is controlled by construction. 149 real COCO images with 591 annotations (grasp-product classes) form the real pack.

Vocabulary. 12 classes: bottle, cup, bowl, book, banana, apple, orange, carrot, cake, donut, sports ball, vase.

Detector. YOLO-World (`yolov8s-world.pt`), imgsz 640, conf 0.15.

Simulator. Execution anchor s_exec = 0.95; phantom rate λ fixed at 0.35 (see Sec. 3.3); 5 seeds averaged.

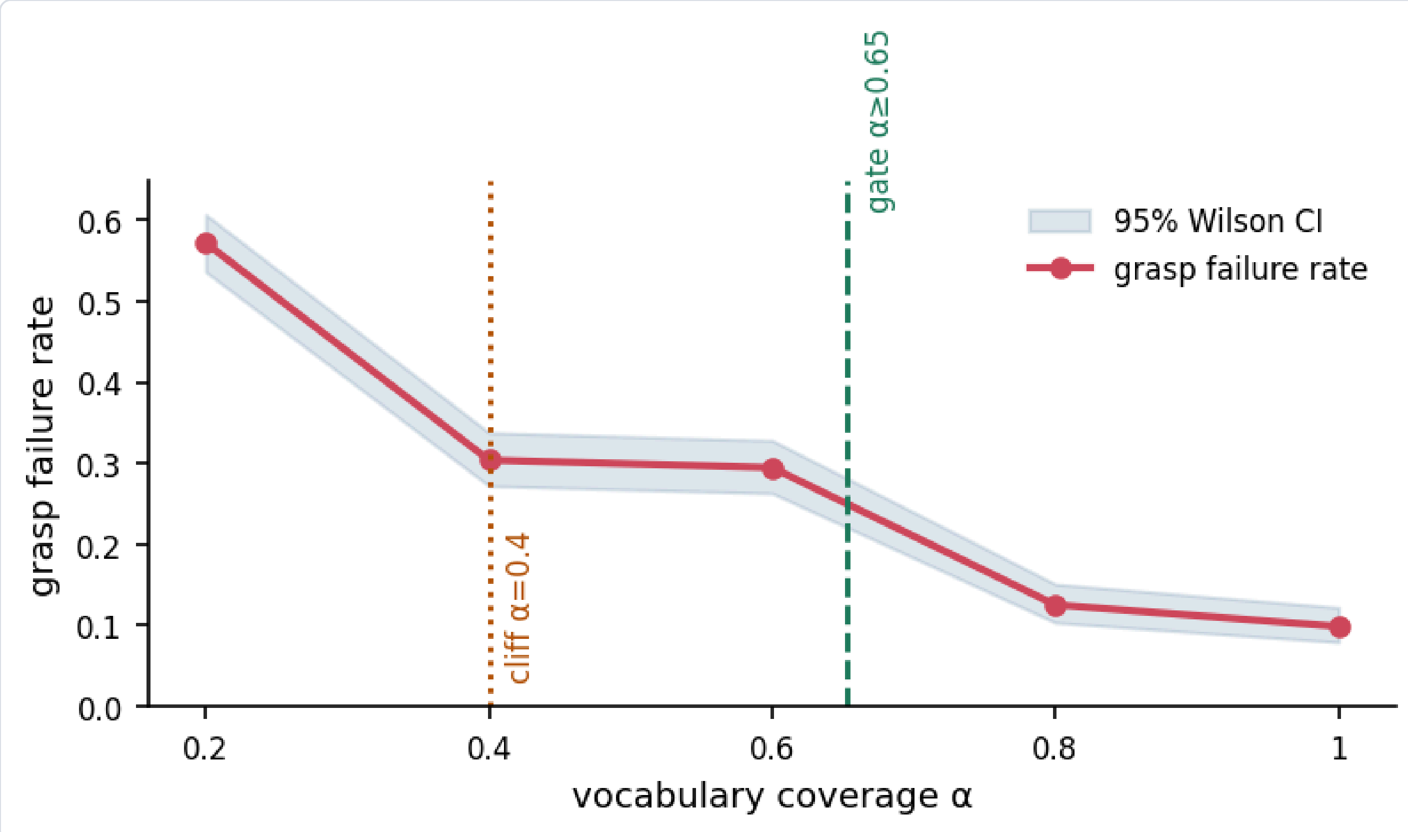
4.2 The perception layer is α-flat

Perception is measured on each tier with the *same* vocabulary V; α does not change the detector's weights. Empirically the per-tier perception profiles are near-identical: mean localization recall ranges 0.40–0.45 across tiers and mean strict recall 0.17–0.23. **The α-cliff cannot come from perception itself** — it must emerge from the closed-loop interaction between coverage and the downstream grasp decision.

4.3 The α-safety cliff

α	grasp failure	95% CI	success
1.0	9.9%	[7.9, 12.2]	90.1%
0.8	12.5%	[10.4, 15.1]	87.5%
0.6	29.5%	[26.3, 32.8]	70.5%
0.4	30.4%	[27.2, 33.8]	69.6%
0.2	**57.2%**	[53.6, 60.7]	42.8%

The frontier is monotone. Adjacent-tier tests: $\alpha = 0.2 \rightarrow 0.4$ (p_Fisher = 0, $q < 10^{-3}$), $\alpha = 0.6 \rightarrow 0.8$ (p_Fisher = 0, $q < 10^{-3}$). The change-point analysis localizes the maximal two-sample separation (statistic 5.5, bootstrap 95% CI [4.5, 6.6]) at $\alpha = 0.4$, our detected cliff. Below it, a 0.2 drop in coverage approximately doubles the failure rate.

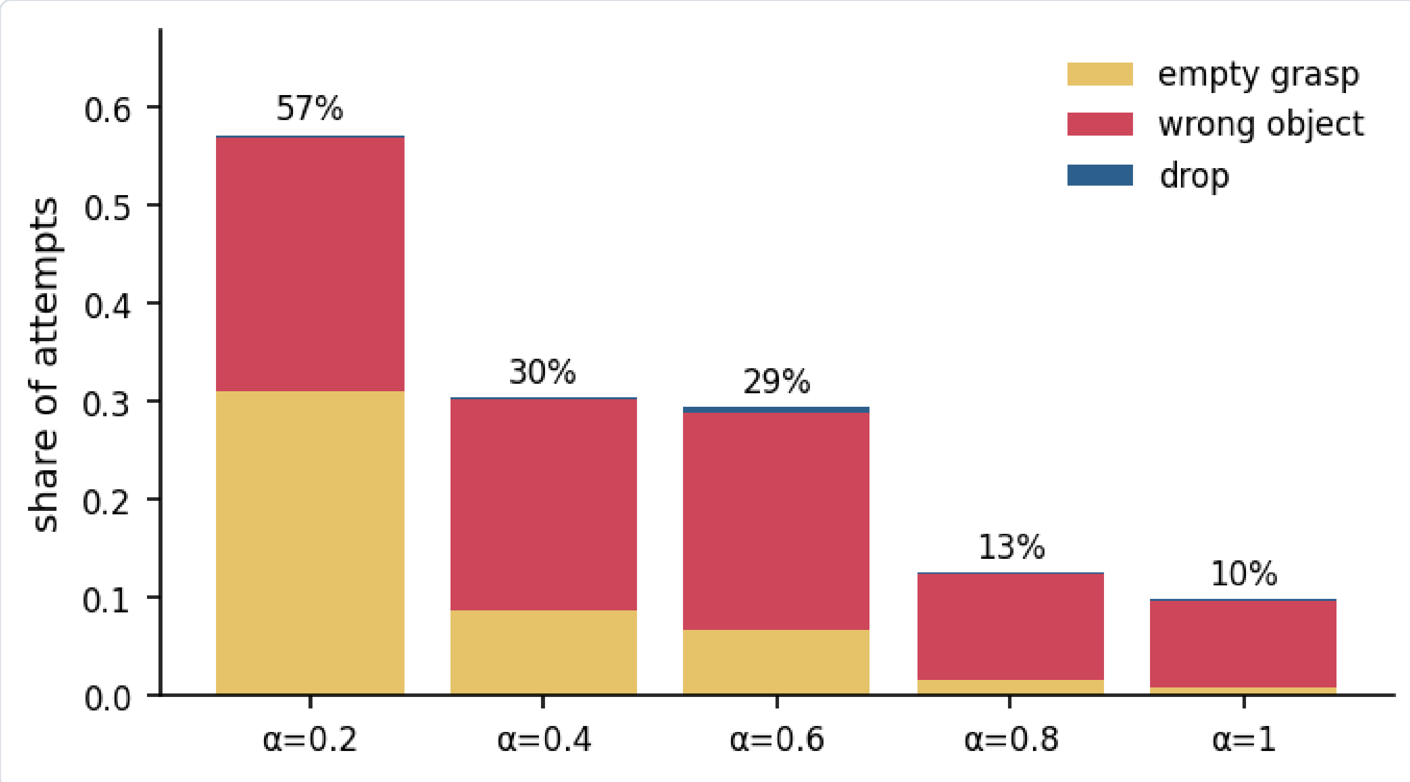


The α-safety cliff: grasp failure rate vs vocabulary coverage, with Wilson 95% CIs, the detected cliff (α=0.4), and the deployment gate (α≥0.653).

4.4 Failure composition

α	empty grasp	wrong object	drop
0.2	31.1%	25.9%	0.3%
0.4	8.7%	21.6%	0.1%
0.6	6.7%	22.1%	0.7%
0.8	1.7%	10.7%	0.1%
1.0	0.9%	8.7%	0.3%

Two regimes are visible. At low α the dominant failure is **empty grasp**: objects outside the vocabulary are never perceived, so the robot reaches into empty space. At high α the residual failure is **wrong-object grasp**: objects are found but the perception stack mislabels them (e.g. a carrot detected as a banana), so the robot grasps the wrong item. These have *different mitigations* — vocabulary expansion vs. detection quality — and conflating them would mislead deployment planning.



Failure composition by coverage: empty grasp dominates at low α, wrong-object grasp at high α.

4.5 The deployment gate

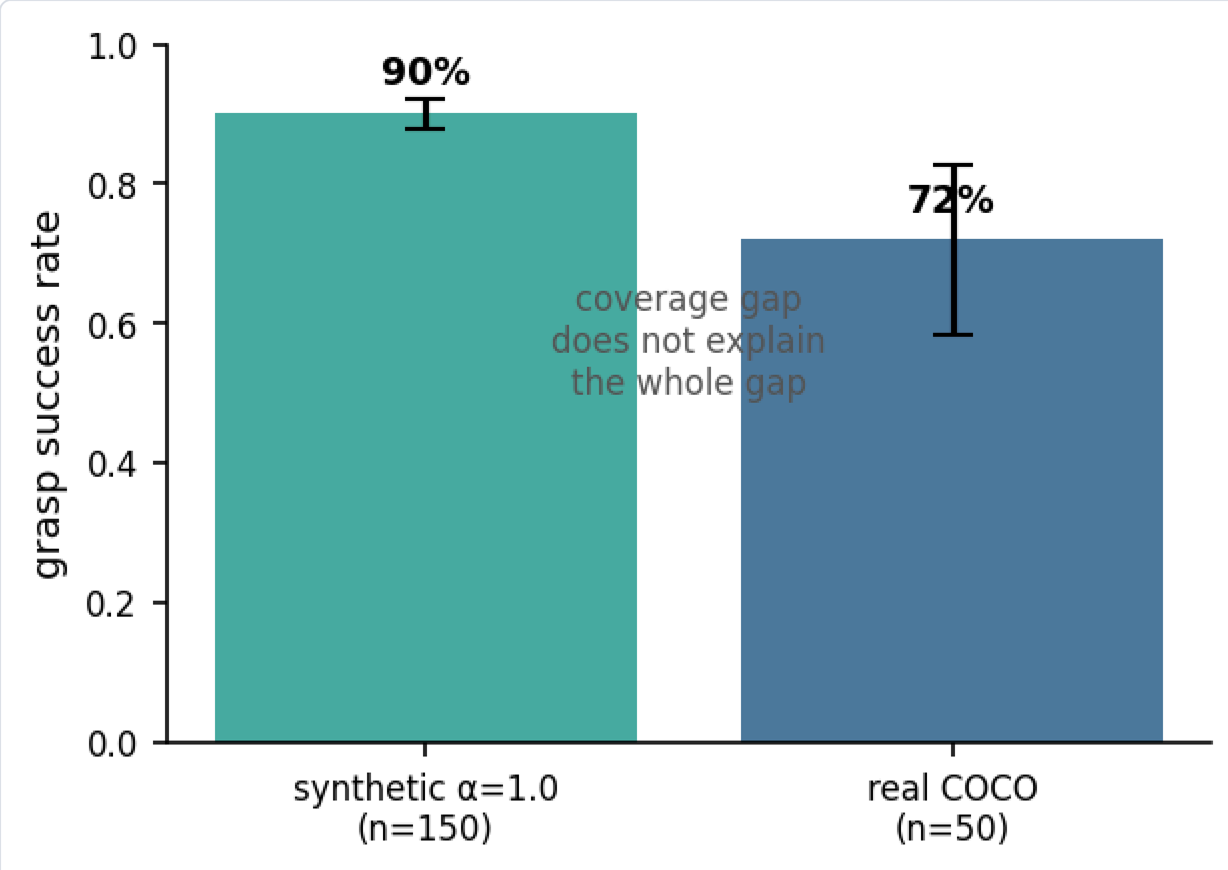
For a target failure bound of 25%, the interpolated gate is

$\alpha^* = 0.653$.

A deployment must ensure that at least 65.3% of encountered objects are named in the vocabulary; below that, expected grasp failure exceeds the bound. The gate interpolates the segment $\alpha = 0.6$ (29.5%) → $\alpha = 0.8$ (12.5%).

4.6 Real-world anchor

Running the same closed-loop pipeline on 50 real COCO product scenes (vocabulary covers the pack by design, so $\alpha = 1.0$) yields **72% success** (95% CI [58.3, 82.5]). This is 18 points below the synthetic $\alpha = 1.0$ level (90.1%). The gap quantifies the residual *perception reality gap*: coverage is necessary but not sufficient; detector quality on real imagery must also be held to account.



Synthetic α=1.0 vs real-scene anchor: coverage is necessary but not sufficient.

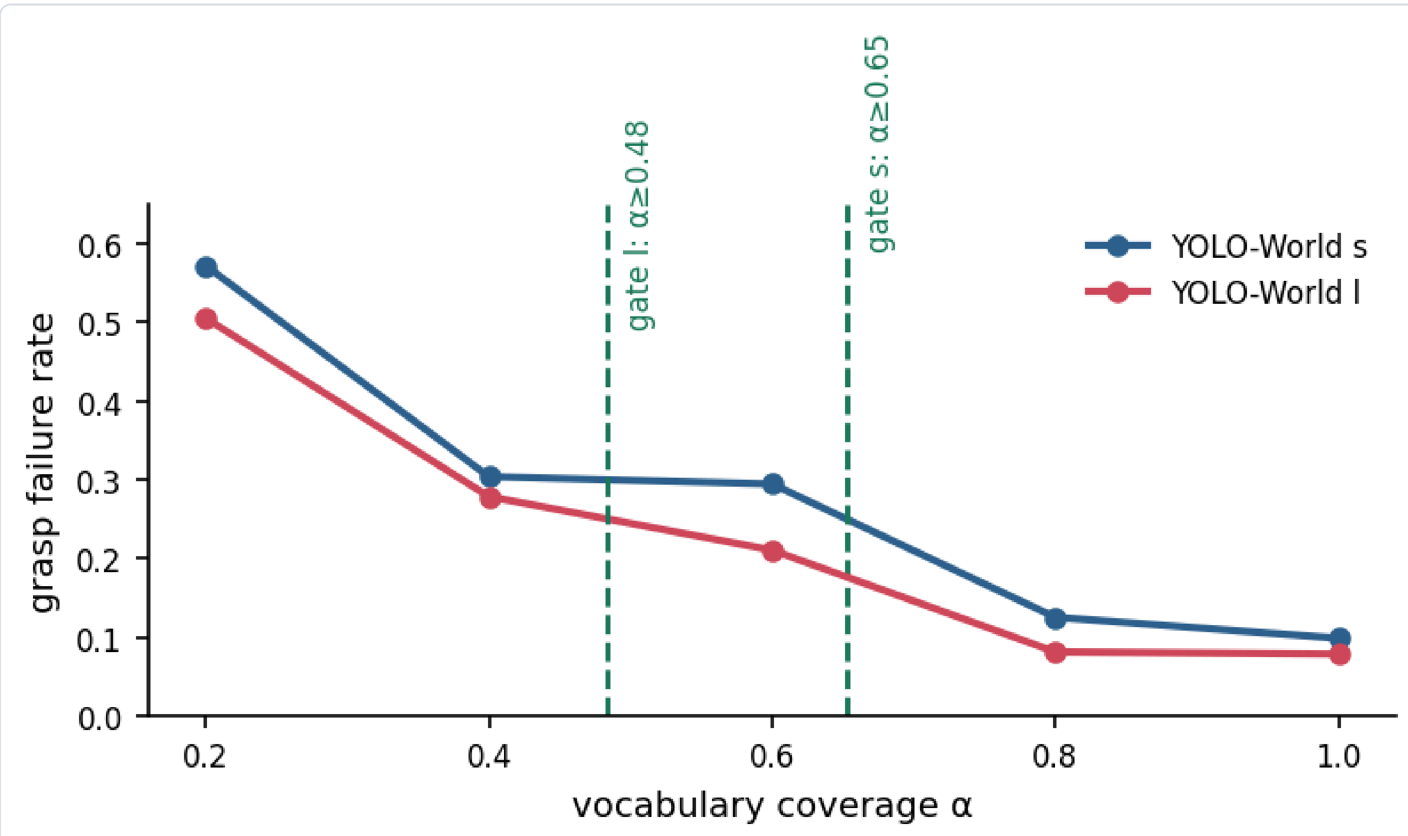
4.7 Detector-scale sensitivity

The audit procedure is deliberately detector-agnostic: the only detector-specific inputs are the measured failure rates. To check whether the *qualitative* law — a coverage cliff with a derived gate — holds as detection quality varies, we repeat the full closed-loop sweep with the larger YOLO-World `yolov8l-world.pt` model on the same corpus and vocabulary. The l model is strictly better at the perception stage — the closed-loop evidence at every tier is uniformly lower failure for l (e.g. α = 1.0: 9.9% → 7.9%; α = 0.2: 57.2% → 50.5%), consistent with improved localization recall — and the comparison uses the same fixed injection parameters (λ = 0.35) with a *more conservative* execution anchor (s_exec = 0.92, vs. 0.95 in the main sweep). We deliberately stress-tested the l arm under a pessimistic execution assumption; since a lower s_exec raises failure at every tier, the l-model gate being *looser* despite this handicap makes the conclusion conservative in the direction we claim.

detector	α	grasp failure	95% CI
s	1.0	9.9%	[7.9, 12.2]
s	0.6	29.5%	[26.3, 32.8]
s	0.4	30.4%	[27.2, 33.8]
s	0.2	57.2%	[53.6, 60.7]

detector	α	grasp failure	95% CI
l	1.0	7.9%	[6.1, 10.0]
l	0.6	21.1%	[18.3, 24.1]
l	0.4	27.9%	[24.8, 31.2]
l	0.2	50.5%	[47.0, 54.1]

Both detectors exhibit a monotone frontier with the cliff fixed at $\alpha = 0.4$ (separation statistic 5.5 for s, 5.7 for l), but the l-model gate is *looser*: $\alpha^* \approx 0.484$ vs. 0.653 for the s model. The form of the law is stable; the numeric gate moves with detection quality, which is exactly the property a deployer needs — better perception pays off as a smaller vocabulary-engineering burden.



Detector-scale sensitivity: the s and l frontiers share the α=0.4 cliff, but the stronger detector moves the gate from 0.653 to 0.484.

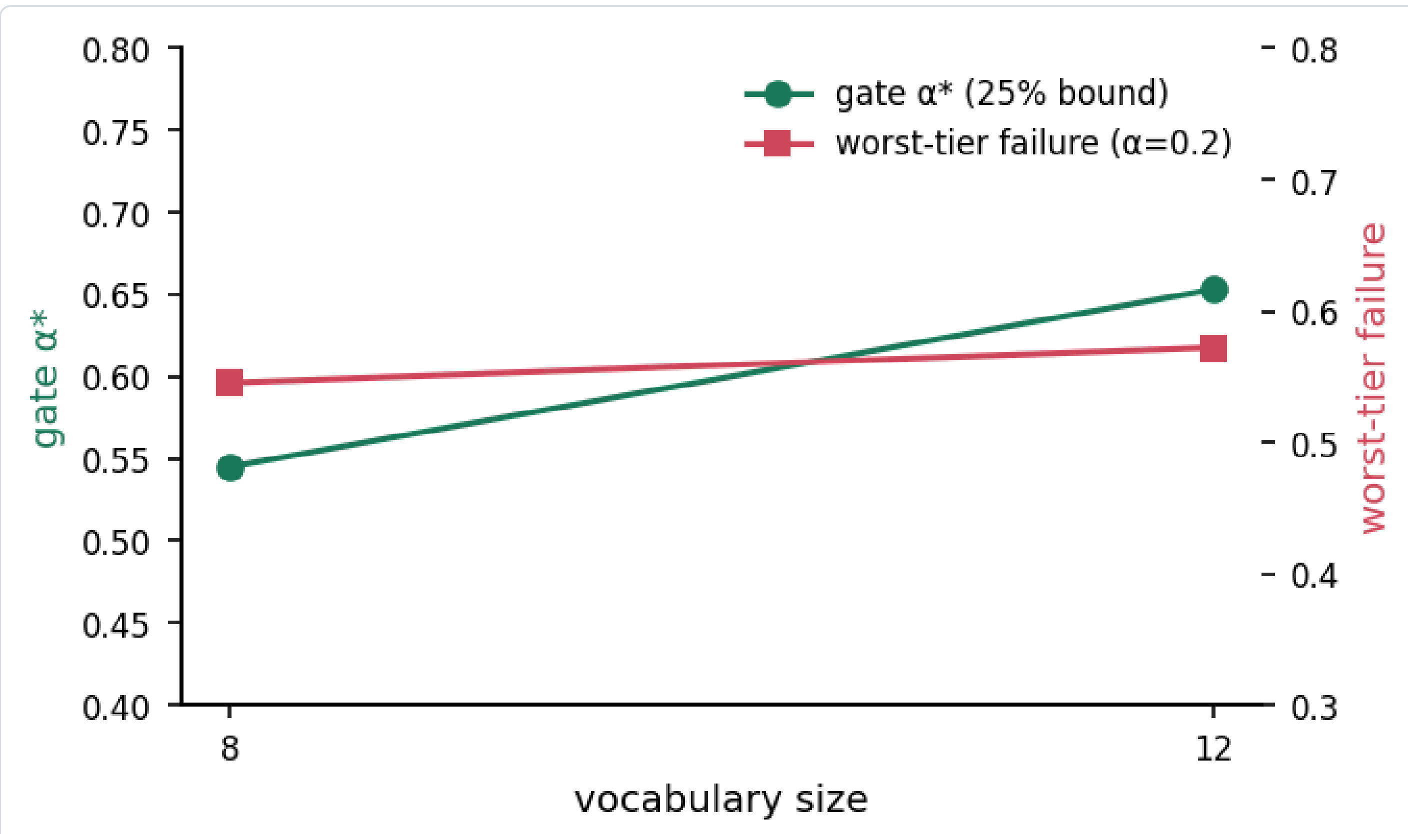
4.8 Vocabulary-size sensitivity

A second question is how the gate responds to the *size* of the vocabulary a deployer maintains, holding detection quality fixed. We rebuild the corpus with a smaller 8-class vocabulary (bottle, cup, bowl, book, banana, apple, orange, carrot) and repeat the audit under the same conservative execution anchor (s_{exec} = 0.92) used for the detector-scale arm. The four dropped classes (cake, donut, sports ball, vase) are the hardest to separate from their neighbours in this catalogue — in the 12-class profiles their mean strict recall (0.10–0.15

across tiers) trails the retained classes (0.18–0.23), and donut is never correctly recognized on the synthetic corpus — so this arm tests whether keeping only a "sharp" vocabulary changes the reliability budget.

vocab size	gate α* (25% bound)	worst-tier failure (α=0.2)
12	0.653	57.2%
8	0.545	54.5%

The 8-class gate is *looser* (0.545 vs. 0.653) and the worst tier is slightly safer (54.5% vs. 57.2%), and both effects hold despite the 8-class arm using the pessimistic execution anchor s_exec = 0.92. The direction is the opposite of a naive "smaller vocabulary ⇒ more coverage pressure" story: removing the four confusable classes reduces per-class confusion (e.g. vase/bottle, cake/donut, and the strongest confuser in this catalogue — everything collapsing onto *book*), so the perception layer is cleaner at every α. The right engineering reading is that the gate is sensitive not just to *how many* classes are maintained but to *which* ones — vocabulary composition, not vocabulary count, drives the coverage budget.



Vocabulary-size sensitivity: gate α* and worst-tier failure for the 8- and 12-class vocabularies.

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5. Limitations

- **Fidelity is relative, not physical.** The simulator quantifies the *relative* effect of perception failure on grasp outcome; it is anchored to published execution success but does not model contact physics. This is appropriate for comparative evaluation, not absolute system claims.

- **Synthetic scenes.** Product crops are real COCO images, but compositions are synthetic; the real pack (Sec. 4.6) partially guards against this.
- **Injection parameters are assumptions, not measurements.** The measured OOV false-positive rate in this corpus is ≈ 0, so the phantom rate $\lambda = 0.35$ is a deliberately conservative stress assumption (Sec. 3.3); likewise the execution anchor s_exec is taken from published grasping success, not from our own robot. We report gates *under these assumptions*; a deployer with a different execution reliability should re-run the closed-loop stage with their own s_exec (the script exposes both as CLI flags).
- **Anchor mismatch across arms.** The sensitivity arms (Secs. 4.7–4.8) were run under the conservative s_exec = 0.92 while the main sweep used 0.95; since a lower execution success only *raises* failure everywhere, the qualitative findings (cliff location, relative gate ordering) are unaffected, and the *direction* of the detector-scale result is conservative. Future runs with identical anchors are supported by the CLI.
- **Two detectors, two vocabularies.** We sweep detector scale and vocabulary size but not architectures; the *procedure* is agnostic, and each new detector or vocabulary is a re-run of the same four stages.

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6. Conclusion

Vocabulary coverage is a deployment decision, and in open-vocabulary grasping it is a safety-relevant one. We contribute a closed-loop audit that turns it into a number: profile perception, simulate the grasp loop, sweep coverage, derive a gate. Applied to our corpus, it yields an α-safety cliff at $\alpha = 0.4$, a deployment gate of $\alpha \geq 0.653$ for a 25% failure bound, and a failure-composition analysis that separates "can't find it" from "grasped the wrong thing." Detector-scale and vocabulary-size sweeps confirm the law's form and give deployers a quantitative budget: stronger perception relaxes the coverage requirement (0.653 → 0.484, and this relaxation is measured under a *more pessimistic* execution anchor for the l model), and vocabulary *composition* — which classes are maintained — matters as much as vocabulary count.

References

1. Deng, S., Yan, M., Wei, S., et al. *GraspVLA: a Grasping Foundation Model Pre-trained on Billion-scale Synthetic Action Data.* CoRL 2025. arXiv:2505.03233.
2. Kim, M. et al. *OpenVLA: An Open-Source Vision-Language-Action Model.* CoRL 2024.
3. Cheng, T. et al. *YOLO-World: Real-Time Open-Vocabulary Object Detection.* CVPR 2024.
4. Lin, T.-Y. et al. *Microsoft COCO: Common Objects in Context.* ECCV 2014.
5. Brohan, A. et al. *RT-1: Robotics Transformer for Real-World Control at Scale.* RSS 2023.
6. Liu, S. et al. *Grounding DINO: Marrying DINO with Grounded Pre-Training.* ICCV 2023.
7. Minderer, M. et al. *Simple Open-Vocabulary Object Detection with Vision Transformers.* ECCV 2022.
8. Gupta, A. et al. *LVIS: A Dataset for Large Vocabulary Instance Segmentation.* CVPR 2019.
9. Liu, B. et al. *LIBERO: Benchmarking Knowledge Transfer for Lifelong Robot Learning.* NeurIPS 2022.

10. James, S. et al. *RLBench: The Robot Learning Benchmark & Learning Environment*. RA-L 2020.
11. Makoviychuk, V. et al. *Isaac Gym: High Performance GPU-Based Physics Simulation for Robot Learning*. NeurIPS 2021 Datasets and Benchmarks.
12. Zhu, Y. et al. *robosuite: A Modular Simulation Framework and Benchmark for Robot Learning*. arXiv:2009.12293, 2020.